

Continuous random variable

A continuous random variable is a random variable that continuous number of values i.e. infinite (uncountable) values in Ω .

probability density function (pdf)

$f(x)$ is said to be a pdf if

- $f(x) \geq 0$;
- $\int_{-\infty}^{\infty} f(x) dx = 1$.

If X has pdf f , then the probability that $X \in B$ is

$$P(X \in B) = \int_{x \in B} f(x) dx.$$

(Interpreted as the area of region under f over B .)

Properties of Probabilities in Continuous r.v.

Given that $F(x) = \int f(x) dx$ (or equivalently $\frac{d}{dx}[F(x)] = F'(x) = f(x)$.)

- $P\{a \leq X \leq b\} = \int_a^b f(x) dx = F(b) - F(a)$.
- $P\{X = a\} = 0$
- $P\{a \leq X \leq b\} = P\{a < X \leq b\} = P\{a \leq X < b\} = P\{a < X < b\}$

Expectation of Continuous r.v. $\mathbb{E}(X)$ and $\mathbb{E}(g(X))$

Given pdf $f(x)$ and real-valued function $g(x)$, the expectation $\mathbb{E}(X)$ and $\mathbb{E}(g(X))$ are defined as

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x) dx \quad \mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x) f(x) dx.$$

if $\int_{-\infty}^{\infty} |x| f(x) dx < \infty$ and $\int_{-\infty}^{\infty} |g(x)| f(x) dx < \infty$ exists respectively.

Properties of Variance are the same as discrete r.v. .

Uniform r.v. , $X \sim \text{Unif}(a, b)$, where $a < b$

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in [a, b] \\ 0 & \text{otherwise.} \end{cases}$$

$$P(X \leq x) = F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } x \in [a, b] \\ 1 & \text{if } x > b. \end{cases}$$

Expectation and Variance of $X \sim \text{Unif}(a, b)$

$$\mathbb{E}(X) = \frac{a+b}{2} \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

Normal Distribution

The density function of the normal distribution depends on two parameters $\mu \in \mathbb{R}$ (mean) and $\sigma > 0$ (standard deviation). $X \sim N(\mu, \sigma^2)$, with pdf

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty.$$

Clearly, $\mathbb{E}(X) = \mu$ and $\text{Var}(X) = \sigma^2$.

Standard Normal distribution: $\mu = 0, \sigma = 1, Z \sim N(0, 1)$.

Define $P(Z \leq a) = \Phi(a)$, where Z follows a Standard Normal distribution.

Symmetrical properties of Normal Distribution: $X \sim N(\mu, \sigma^2)$

- $P(X \leq \mu - a) = P(X \geq \mu + a) = 1 - P(X \leq \mu + a)$
- X can be standardised into a Standard Normal distribution by defining

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

- Useful for reading off z -values:

$$P(Z \leq a) = P(Z \geq -a) = 1 - P(Z \leq -a) \text{ or } \Phi(a) = 1 - \Phi(-a).$$

De-Moivre Laplace Limit Theorem

Let p be a constant and $S_n \sim \text{Bin}(n, p)$, then for any $a < b$,

$$\lim_{n \rightarrow \infty} \left\{ a \leq \frac{S_n - np}{\sqrt{np(1-p)}} \leq b \right\} = \Phi(b) - \Phi(a)$$

Normal Approximation to Binomial Distribution: $np(1-p) \geq 10$.

Note that Binomial is a discrete r.v. , so the following changes would be needed to approximate $X \sim \text{Bin}(n, p)$

- $P(X = i) \stackrel{\text{approximation}}{=} P\left(i - \frac{1}{2} < X < i + \frac{1}{2}\right).$
- $P(X \leq i) \stackrel{\text{approximation}}{=} P\left(X \leq i + \frac{1}{2}\right).$
- $P(X < i) \stackrel{\text{approximation}}{=} P\left(X \leq i - \frac{1}{2}\right).$

Exponential Distribution - $X \sim \text{Exponential}(\lambda)$

Used to model between the occurrence of events in an interval of time. i.e. Queuing Scenarios. Given parameter $\lambda > 0$,

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$$

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}, x \geq 0.$$

Expectation and Variance of $X \sim \text{Exponential}(\lambda)$

$$\mathbb{E}(X) = \frac{1}{\lambda} \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$